NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-27/69A
18 May 1977
SUPERSEDING
MIL-T-27/69
14 April 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY TF5S21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.

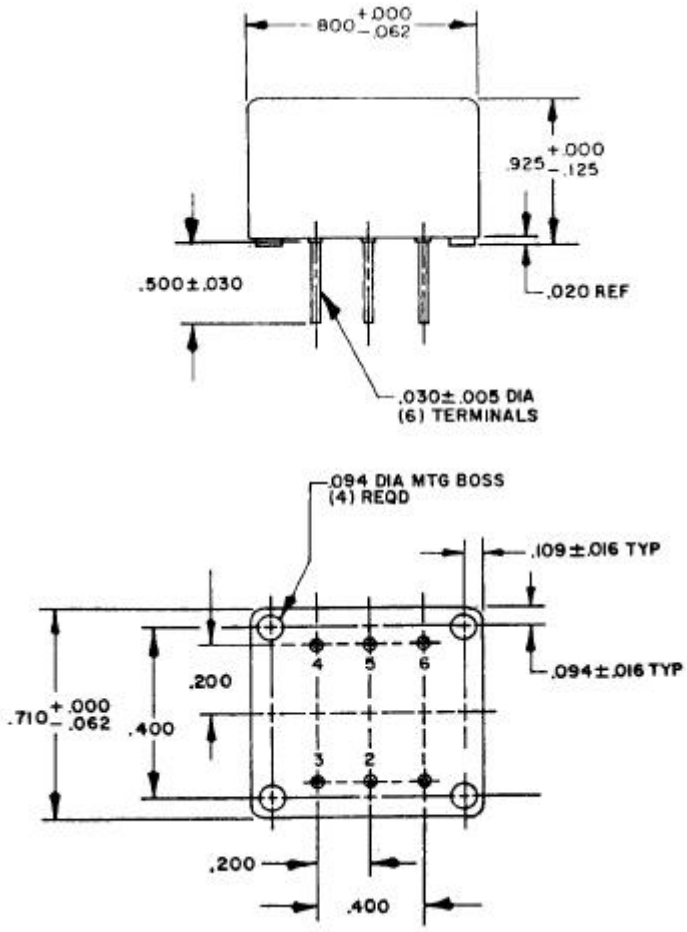
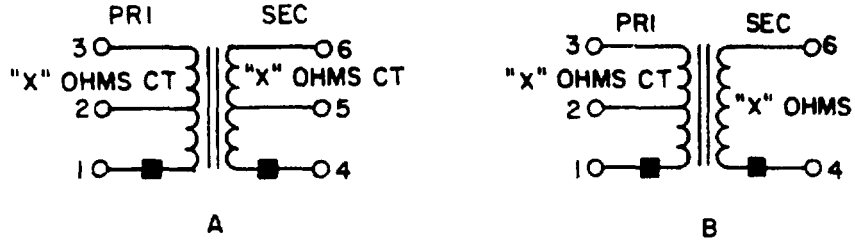


FIGURE 1. Dimensions and configuration.

FSC 5950

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INCHES	MM	INCHES	MM
005	.13	125	3 18
016	41	200	5 08
020	51	400	10 16
030	76	500	12 70
062	1 57	710	18 03
094	2 39	800	20 32
109	2 77	925	23 50



WORKING VOLTAGE. 175 V
ALTITUDE . 70 kFT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAMS AND MARKING

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3 Unless otherwise specified, tolerance is ± 016 (41 mm)
4 Marking shall be on sides and/or top of case
5 Electrical values shall be marked as specified in table I, as applicable

FIGURE 1 Dimensions and configuration - Continued

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.
Power level: 500 milliwatts maximum at 200 hertz
Working voltage 175 volts.
Frequency range: 200 hertz to 20 kilohertz.

TABLE I. Electrical ratings.1/

Dash No.	Circuit diagram (see figure 1)	Primary impedance ohms (1-3)	Secondary impedance ohms (4-6)	Primary DCR ohms ±25%	Secondary DCR ohms ±25%	Primary unbalanced DC current max mA
01	A	25 k CT	1,000 CT	2,700	150	0
02	A	10 k CT	10 k CT	1,100	1,200	1.0
03	A	10 k CT	1,500 CT	1,100	180	1.0
04	A	10 k CT	600 CT	1,100	75	1.0
05	B	10 k CT	4	1,100	0.5	1.0
06	A	4,000 CT	10 k CT	440	1,200	1.5
07	B	1,500 CT	15	170	1.8	2.5
08	A	1 k CT	600 CT	105	75	3.0
09	B	1 k CT	4	105	0.5	3.0
10	A	600 CT	25 k CT	65	3,700	4.0
11	A	600 CT	600 CT	65	75	4.0
12	B	600 CT	60	65	7	2.0
13	B	600 CT	15	65	2	2.0
14	B	600 CT	4	65	0.5	2.0
15	A	250 CT	600 CT	30	75	2.0

1/ Qualification testing and approval to M27/69-01 shall be sufficient to grant qualification approval to M27/69-02 through M27/69-15.

Design and construction:
Dimensions and configuration. See figure 1.
Duty cycle: Continuous.
Case: Encapsulated.
Material: Plastic.
Terminals Pin (printed circuit).
Material Nickel iron alloy, tin plated
Diameter: 0.030 ±0.005 inch
Length 0.500 ±0 030 inch.
Weight 26.93 grams maximum
Altitude 70,000 feet maximum.
Operating temperature range -55°C to +130°C.

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage
At sea level. 500 volts rms.
At barometric pressure 300 volts rms.

Electrical characteristics See table II
Frequency response
Reference frequency: 1 kilohertz.
Polarity Additive, with terminals 3 and 4 connected.

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TABLE II. Electrical characteristics.

Dash No.	Frequency response at 200 Hz to 20 kHz ±1 dB			
	Z_s (1-3) ohms	Z_L (4-6) ohms	E_L volts	Primary DC current (1-3) mA
01	25 k	1,000	3.0	0
02	10 k	10 k	8.0	0.5
03	10 k	1,500	8 0	0.5
04	10 k	600	8.0	0.5
05	10 k	4	8.0	0 5
06	4,000	10 k	5.0	0.75
07	1,500	15	3.0	1 25
08	1 k	600	3.0	1.5
09	1 k	4	3.0	1.5
10	600	25 k	2.0	2.0
11	600	600	2.0	2.0
12	600	60	2.0	2 0
13	600	15	2.0	2 0
14	600	4	2.0	2 0
15	250	600	2 0	3.5

Marking location See figure 1

Part number M27/69-(dash number from table I)

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - MU, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity
Army - EL

Agent
DLA - ES

(Project 5950-0543-3)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-27/69A TRANSFORMERS, AUDIO FREQUENCY, TF5S21ZZ

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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